



Certificate

We hereby confirm that

Muhammad Hamza Latif

has completed the intensive program

AI Project Manager

with 810 hours of programming practice (corresponds to 1080 lessons) successfully.

The course has been delivered remotely.

Participation from 15.9.2025 until 30.01.2026

Filip Vukovic
Program Manager

Petra Kischkewitz
Managing Director

Anastasia Mikheeva
Lead Coach



Certified by

>> neue fische | SPICED

Curriculum AI Project Management

In 720 lessons the students have learned, discussed and practiced the following content in several tasks and projects:

Programming & Database Python

- pandas
- scikit-learn
- TensorFlow/Keras
- Visualization
- Matplotlib
- Seaborn
- SQL
- Unix
- Git
- GitHub

Exploratory Data Analysis EDA

- Data Analysis & Statistics
- Distributional Analysis
- General Statistics
- Data Visualization
- Distribution Plots
- Relationship Plots
- Geo Visualizations

Data Science Portfolio Projects

- EDA Project
- Predictive Modeling Project
- Final Project / Capstone (4 weeks)

Machine Learning Algorithms

- Supervised Learning
- Linear Regression
- Logistic Regression
- KNN
- Decision Trees
- Random Forest
- Ensemble Methods

Deep Learning

- Artificial Neural Networks
- Convolutional Neural Networks

Natural language processing

- Bag of words
- Word embeddings

Large language models

- Prompt Engineering
- Retrieval augmented generation
- AI Agents

Evaluation & Performance Metrics

- Confusion Matrix
- Regression and Classification Metrics
- Error Analysis
- Model Tuning
- Bias-Variance Tradeoff
- Regularization
- Cross-Validation

Optimization

- Gradient Descent
- Loss and Cost Functions

APIs & Web Scraping

Streamlit

Machine Learning Engineering

- Sklearn Pipeline
- MLflow
- Docker and APIs
- Model Deployment and Monitoring

Statistics

- A/B testing

Project management

- AI PM Basics
- Agile Frameworks
- Tools (Jira, Confluence)
- Rituals (Meetings, Refinement)
- Artifacts (DoR, DoD, Stories)
- Request for Proposal
- Feasibility study
- Data product scoping
- Product discovery
- User research
- Prototyping
- Roadmap creation
- Software architecture
- AI/ML architecture

Collaborative Working & Social Learning

- Pair Coding
- Agile Workflow
- Daily Stand-Ups
- Retrospective
- Kanban Project Board
- Daily Class Review
- Team Work & Self-Organization Skills
- Career Coaching & Mentoring

Communication with Stakeholders

- Business Presentation
- Non-Technical Stakeholders
- Data Science Lifecycle Presentation
- Technical Stakeholders
- Stakeholder Review
- Compact & Time-Critical Project Alignment